



# Ethernet Switching Basics



# Foreword

- Data transmission on networks must comply with certain standards. Ethernet protocols define how data frames are transmitted over an Ethernet network. Understanding Ethernet protocols is the basis for fully understanding communication at the data link layer. An Ethernet switch is the main device for implementing data link layer communication. It is essential to understand how an Ethernet switch works.
- This course describes the concepts related to Ethernet protocols, MAC address types, and working process and mechanism of Layer 2 switches.



# Objectives

- On completion of this course, you will be able to:
  - Describe the basic concepts of an Ethernet network.
  - Distinguish MAC address types.
  - Get familiar with the working process of a Layer 2 switch.
  - Get familiar with the structure and generation process of a MAC address table.



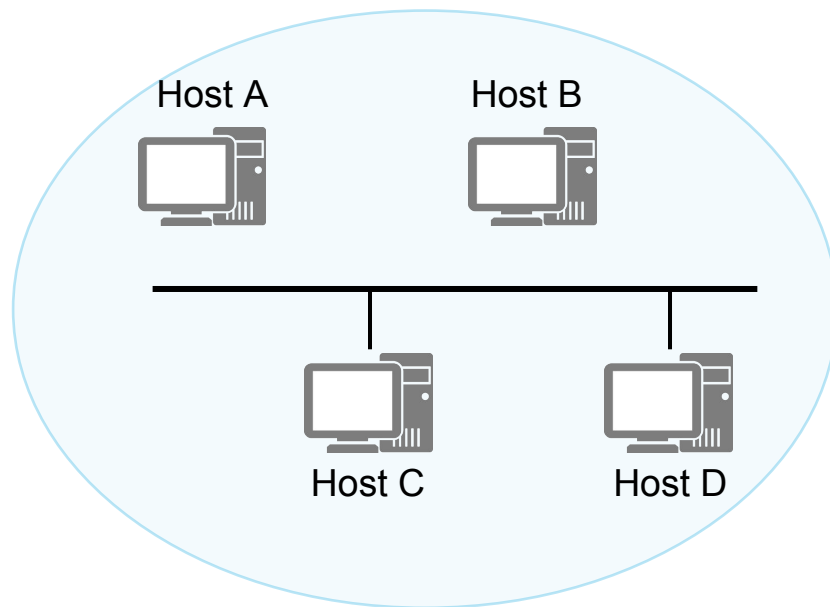
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- 1. Overview of Ethernet Protocols**
2. Overview of Ethernet Frames
3. Overview of Ethernet Switches
4. Process of Data Communication Within a Network Segment

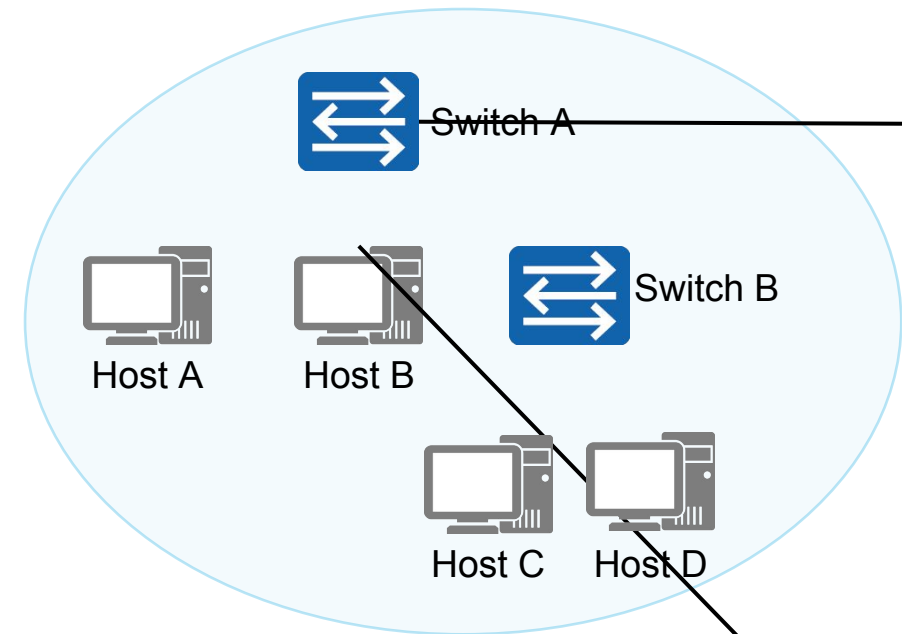
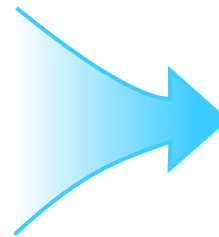


# Ethernet Protocols

- Ethernet is the most common communication protocol standard used by existing local area networks (LANs). It defines the cable types and signal processing methods that are used on a LAN.
- An Ethernet network is a broadcast network built based on the carrier sense multiple access/collision detection (CSMA/CD) mechanism.



**Early Ethernet**

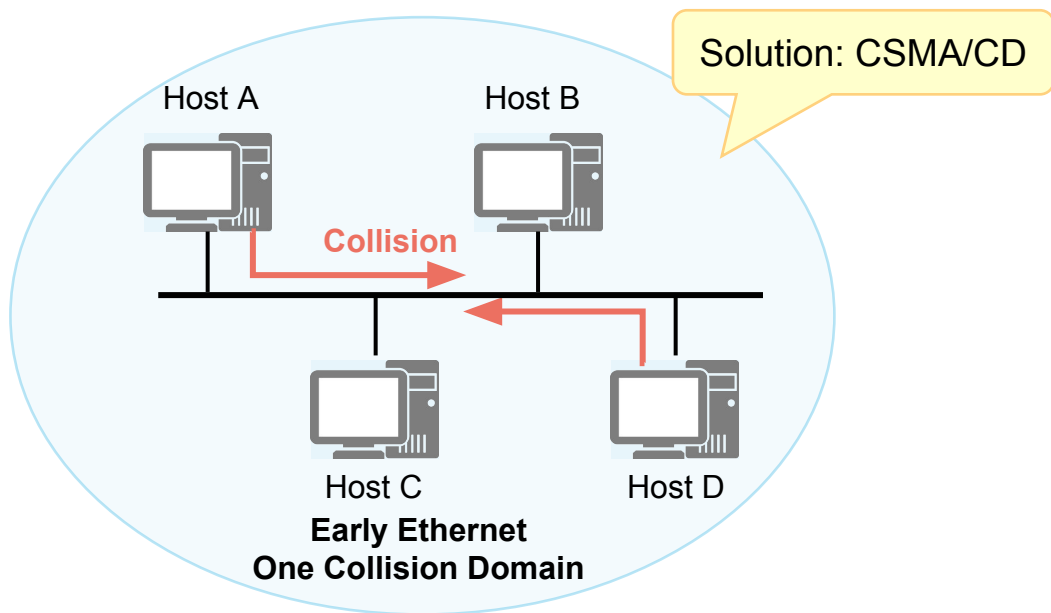


**Switch Networking**

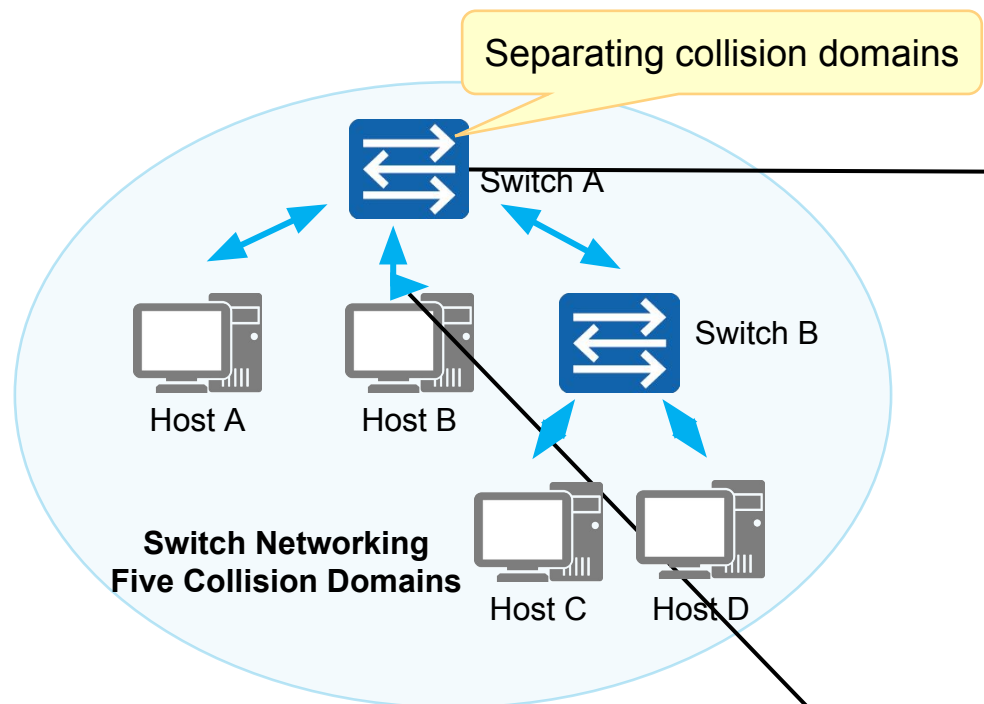


# Collision Domain

- A collision domain is a set of nodes connected to the same shared medium. All nodes in a collision domain compete for the same bandwidth. Packets (unicast, multicast, or broadcast) sent by a node can be received by other nodes.



- On a traditional Ethernet network, multiple nodes on the same medium share the link bandwidth and compete for the right to use the link. As a result, collision occurs.
- The probability that collision occurs increases when more nodes are deployed on a shared medium.

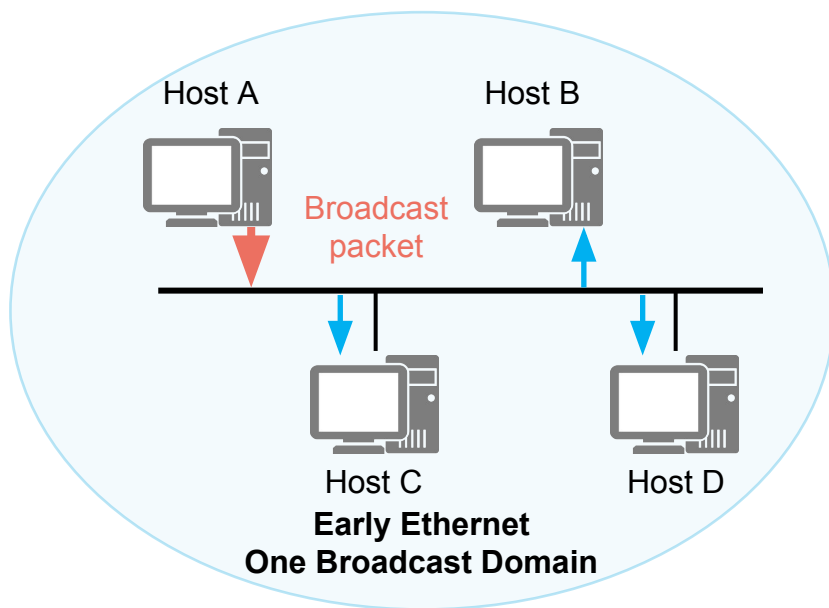


The switch interfaces used to send and receive data are independent of each other and belong to different collision domains. Therefore, collisions do not occur between hosts (or networks) connected through switch interfaces.

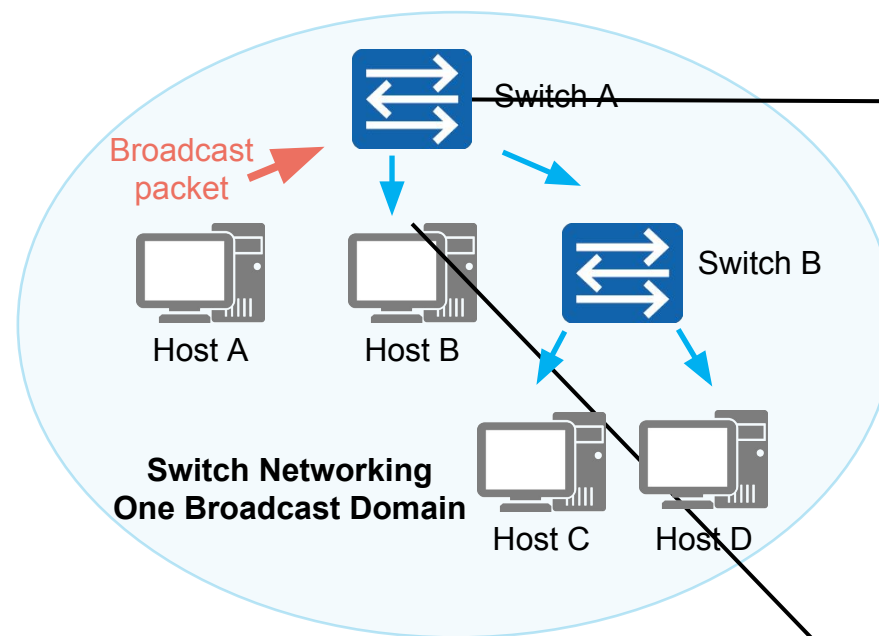


# Broadcast Domain

- The entire access scope of broadcast packets is called a Layer 2 broadcast domain, which is also called a broadcast domain. All hosts in the same broadcast domain can receive broadcast packets.



On a traditional Ethernet network, multiple nodes on the same medium share a link. The broadcast packets sent by a device can be received by all the other devices.



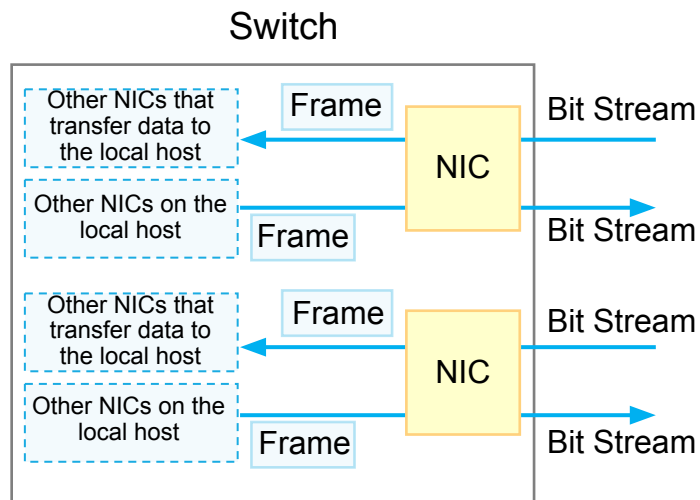
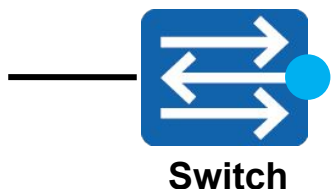
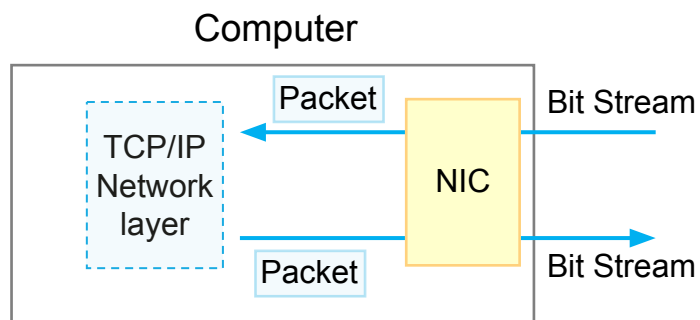
A switch forwards broadcast packets to all interfaces. Therefore, the nodes connected to all interfaces of the switch belong to the same broadcast domain.





# Ethernet NIC

- A network interface card (NIC) is a key component that connects a network device (such as a computer, a switch, or a router) to an external network.



- **Network Port**

- A network port is also called a network interface, interface, or port.

- **NIC**

- Each network port corresponds to a NIC.
- A computer or switch forwards data through a NIC.





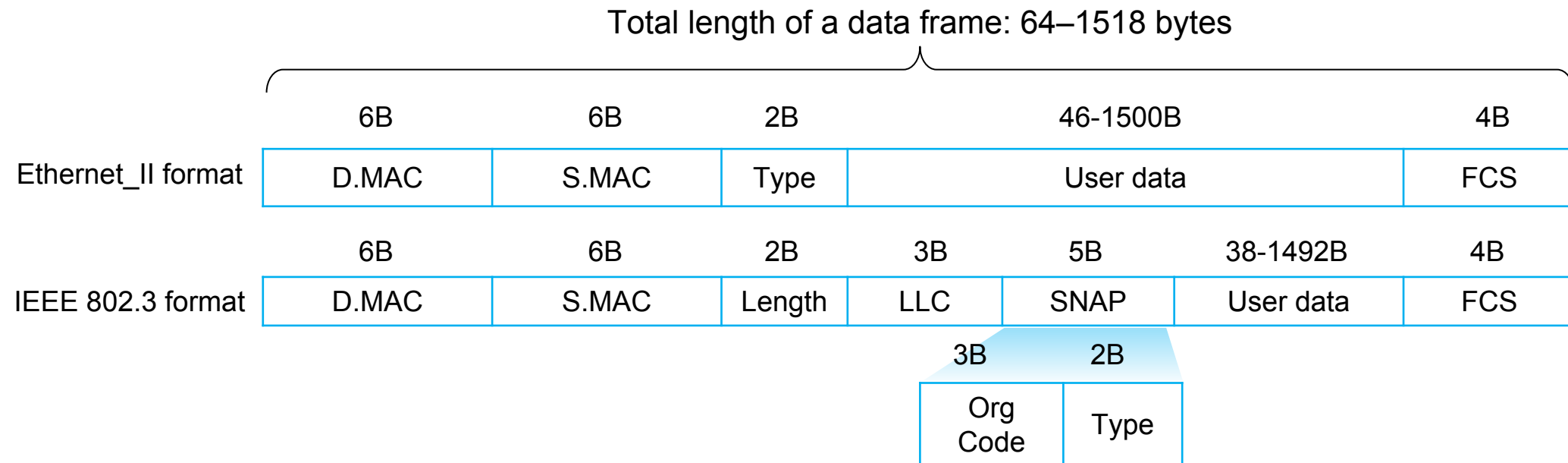
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# Ethernet Frame Format

- The frames used by Ethernet technology are referred to as Ethernet frames.
- Ethernet frames are in two formats: Ethernet\_II and IEEE 802.3.

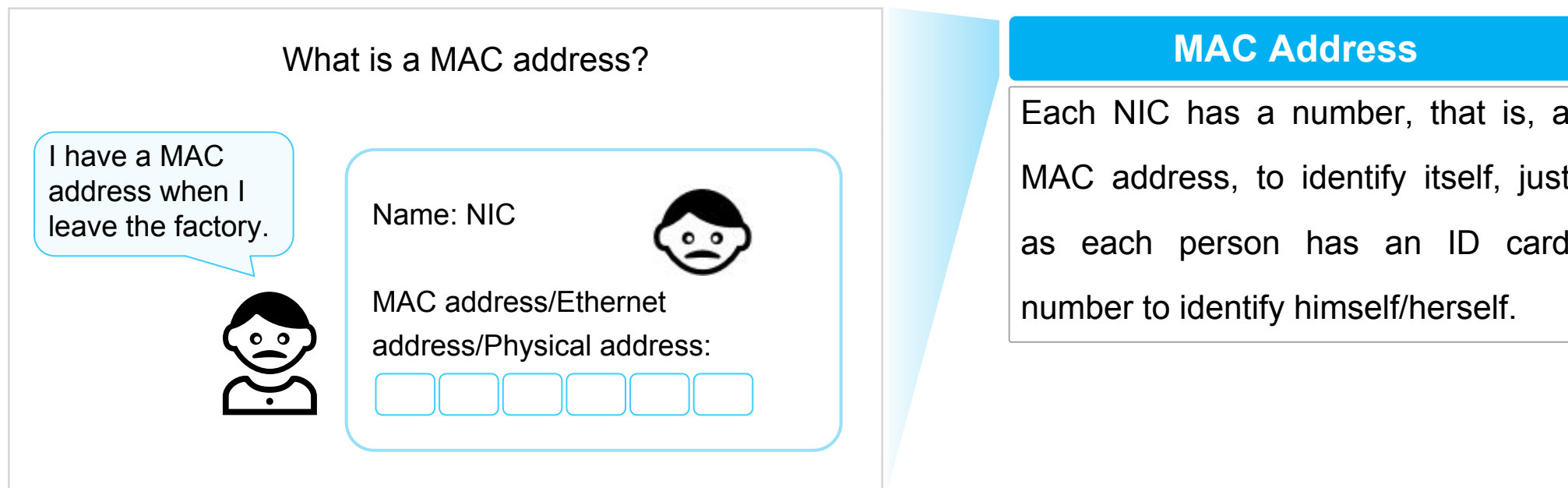






# What Is a MAC Address?

- A media access control (MAC) address uniquely identifies a NIC on a network. Each NIC must have a globally unique MAC address.

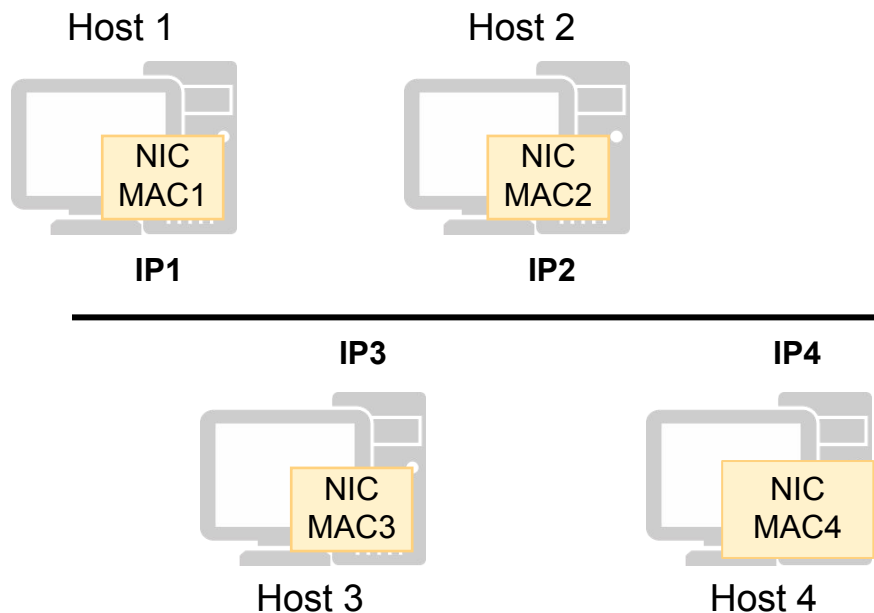




# IP Address Vs. MAC Address

- Each Ethernet device has a unique MAC address before delivery. When the device accesses the network, it assigns an IP address to each host. Why?

## Ethernet



### Characteristics of IP addresses:

- IP addresses are unique.
- IP addresses are **changeable**.
- IP addresses are assigned based on **network topology**.

### Characteristics of MAC addresses:

- MAC addresses are unique.
- MAC addresses **cannot be changed**.
- MAC addresses are assigned based on the **manufacturer**.

Can a network device have either a MAC address or an IP address?





# MAC Address Presentation

- A MAC address is 48 bits (6 bytes) in length.
- As typically represented, MAC addresses are recognizable as six groups of two hexadecimal digits, separated by hyphens, colons, or without a separator.

For example, 00-1E-10-DD-DD-02 or 001E-10DD-DD02

|             |           |           |           |           |           |           |        |
|-------------|-----------|-----------|-----------|-----------|-----------|-----------|--------|
| Hexadecimal | 00        | 1E        | 10        | DD        | DD        | 02        | 6-byte |
| Binary      | 0000 0000 | 0001 1110 | 0001 0000 | 1101 1101 | 1101 1101 | 0000 0010 | 48-bit |

Conversion  
between  
hexadecimal and  
binary digits

Power

|       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|
| $2^3$ | $2^2$ | $2^1$ | $2^0$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
| 8     | 4     | 2     | 1     | 8     | 4     | 2     | 1     |
| 0     | 0     | 0     | 1     | 1     | 1     | 1     | 0     |

= 1

= 8+4+2=14=E

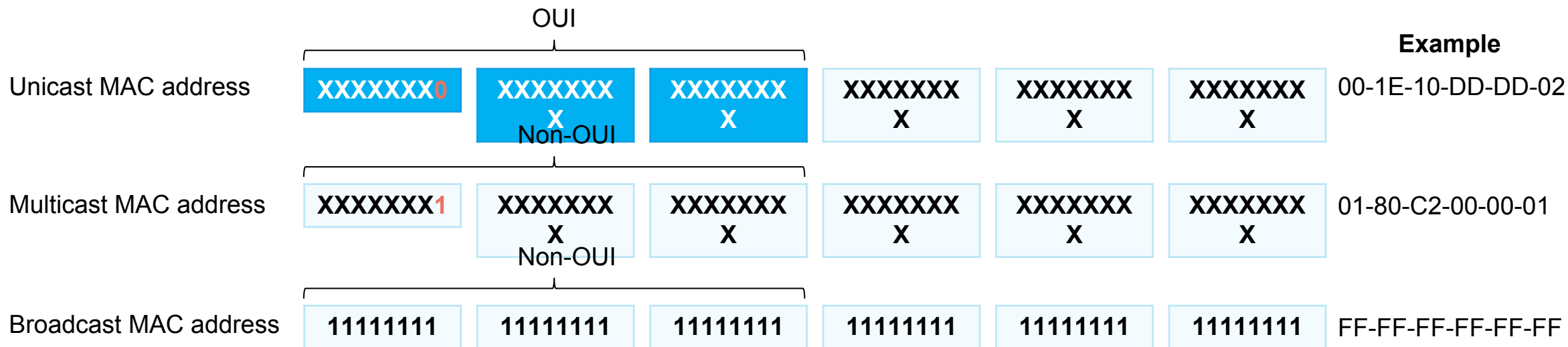


# MAC Address Composition and Classification

- Organizationally unique identifier (OUI): a 24-bit (3-byte) number. It is a globally unique identifier assigned by the IEEE.
- Company ID (CID): a 24-bit (3-byte) number. It is assigned by a manufacturer.



- MAC address classification:









# Unicast Ethernet Frame

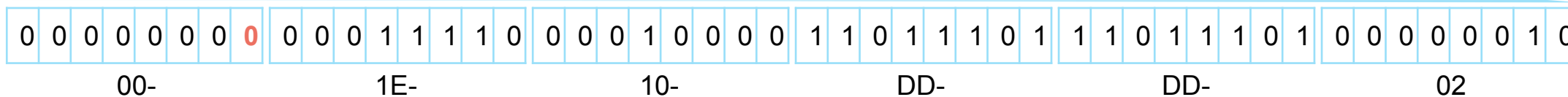
- A unicast Ethernet frame is also called a unicast frame.
- The destination MAC address of a unicast frame is a unicast MAC address.



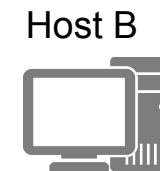
**D.MAC:**  
**00-1E-10-DD-DD-02**

S.MAC

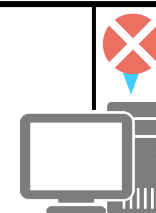
DATA



Unicast  
Frame



Host C

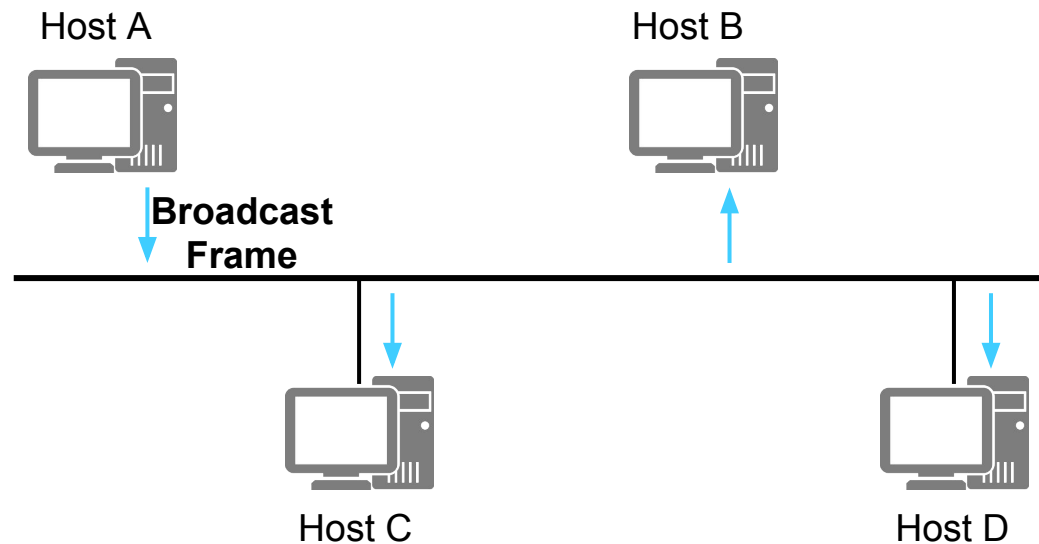
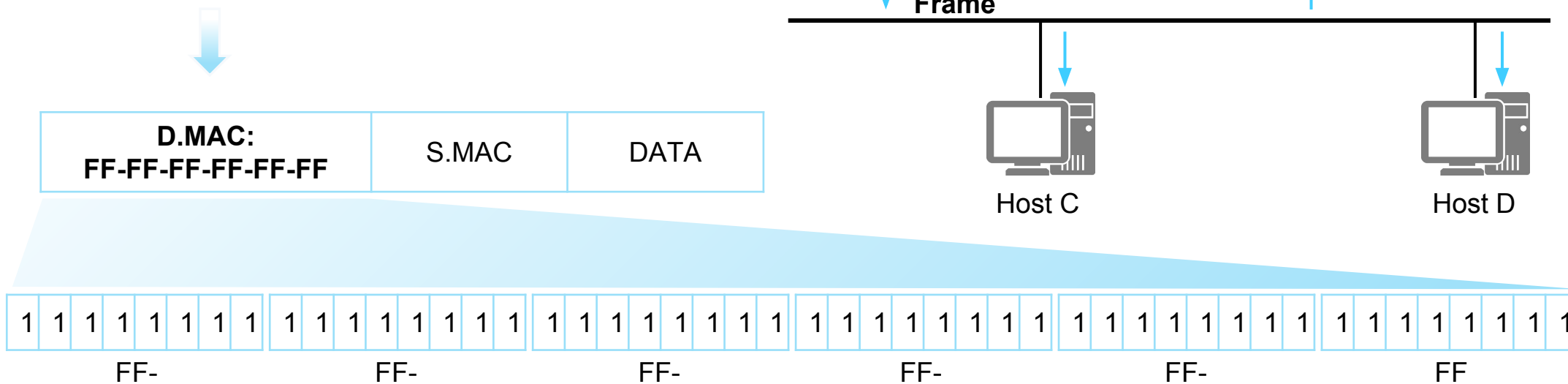


Host D



# Broadcast Ethernet Frame

- A broadcast Ethernet frame is also called a broadcast frame.
- The destination MAC address of a broadcast frame is a broadcast MAC address.





# Multicast Ethernet Frame

- A multicast Ethernet frame is also called a multicast frame.
- The destination MAC address of a multicast frame is a unicast MAC address.

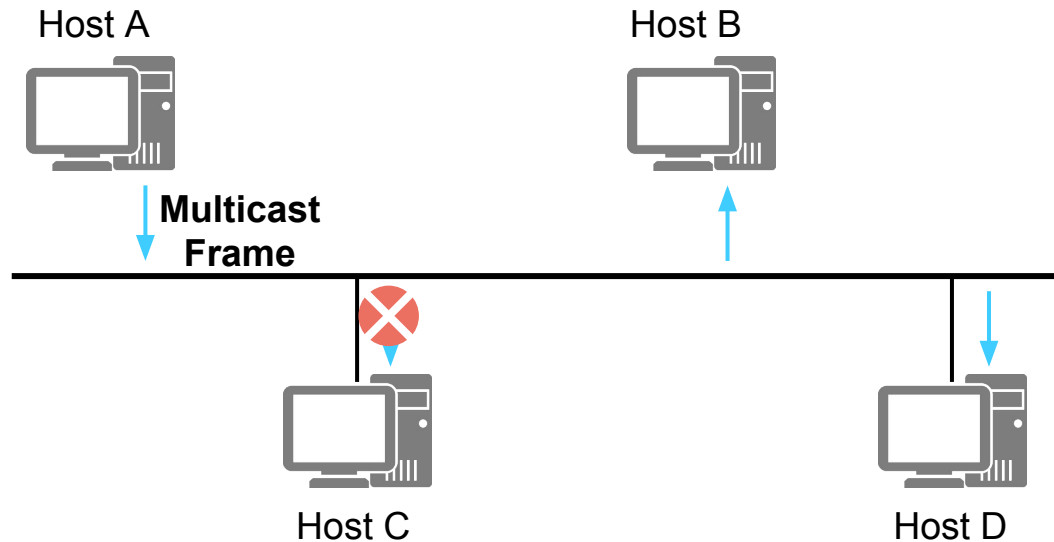
**D.MAC:**  
**01-80-C2-00-00-01**

S.MAC

DATA

0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 1 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1

01- 80- C2- 00- 00- 01



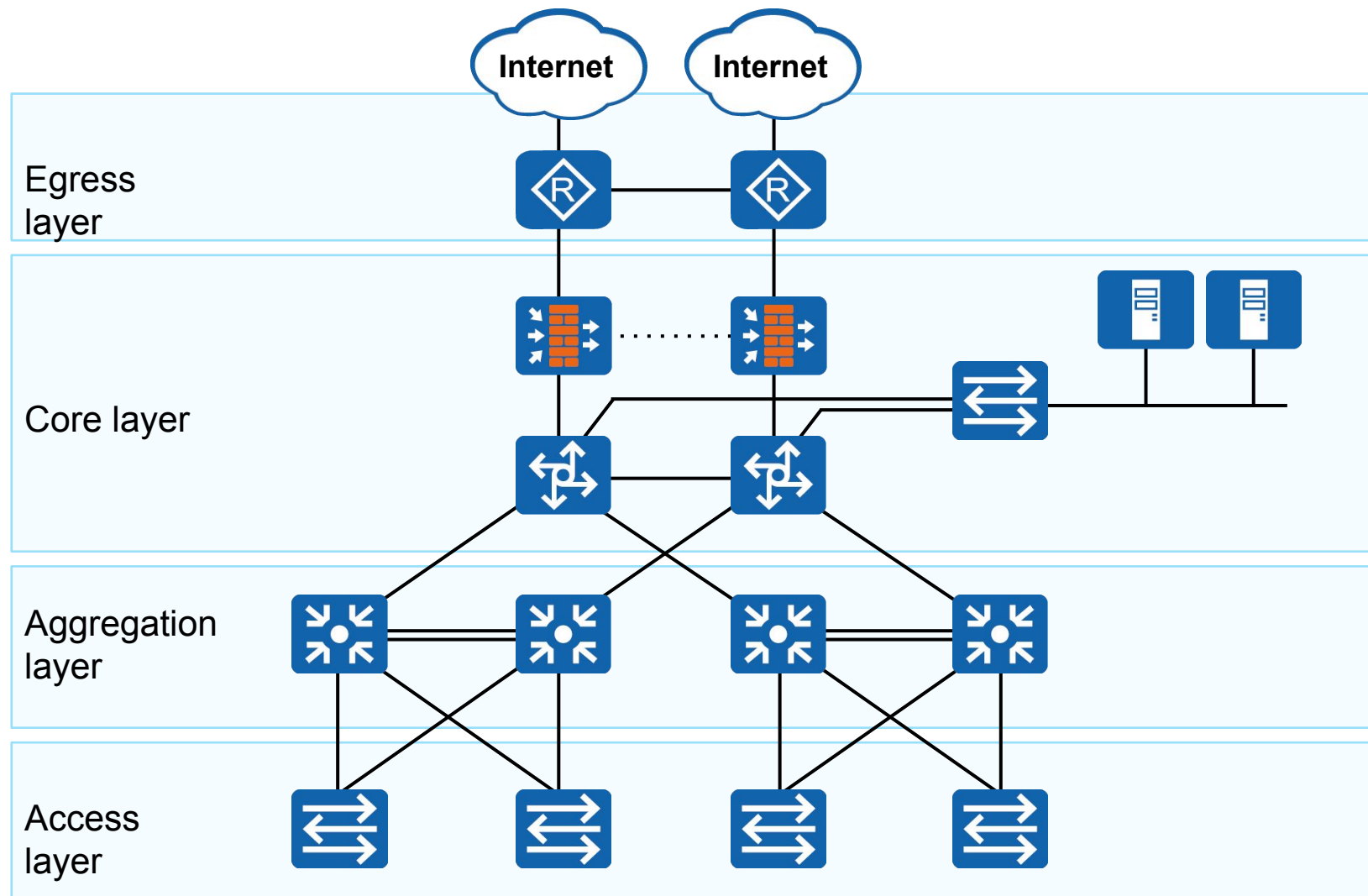


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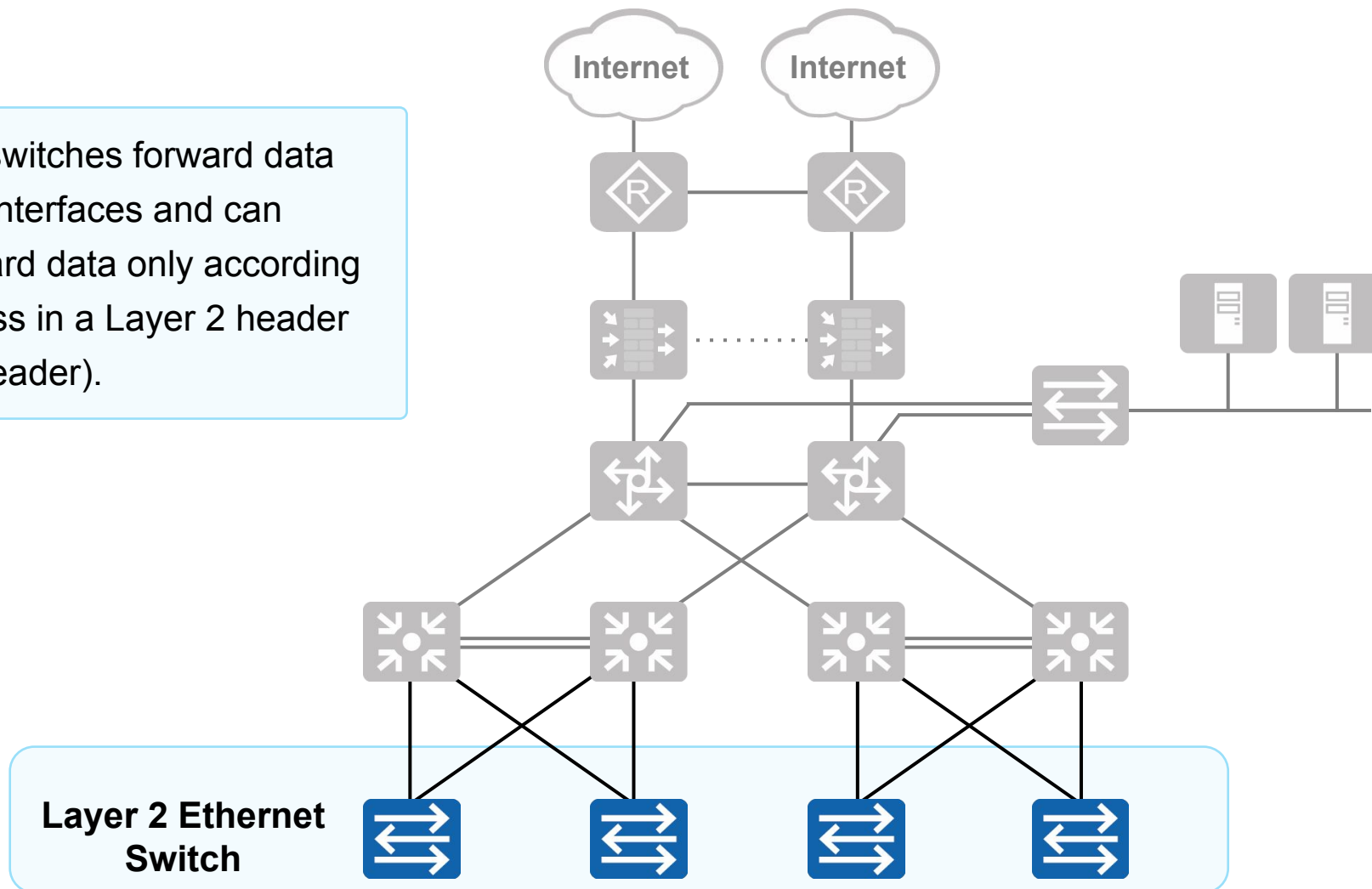
# Typical Architecture of a Campus Network





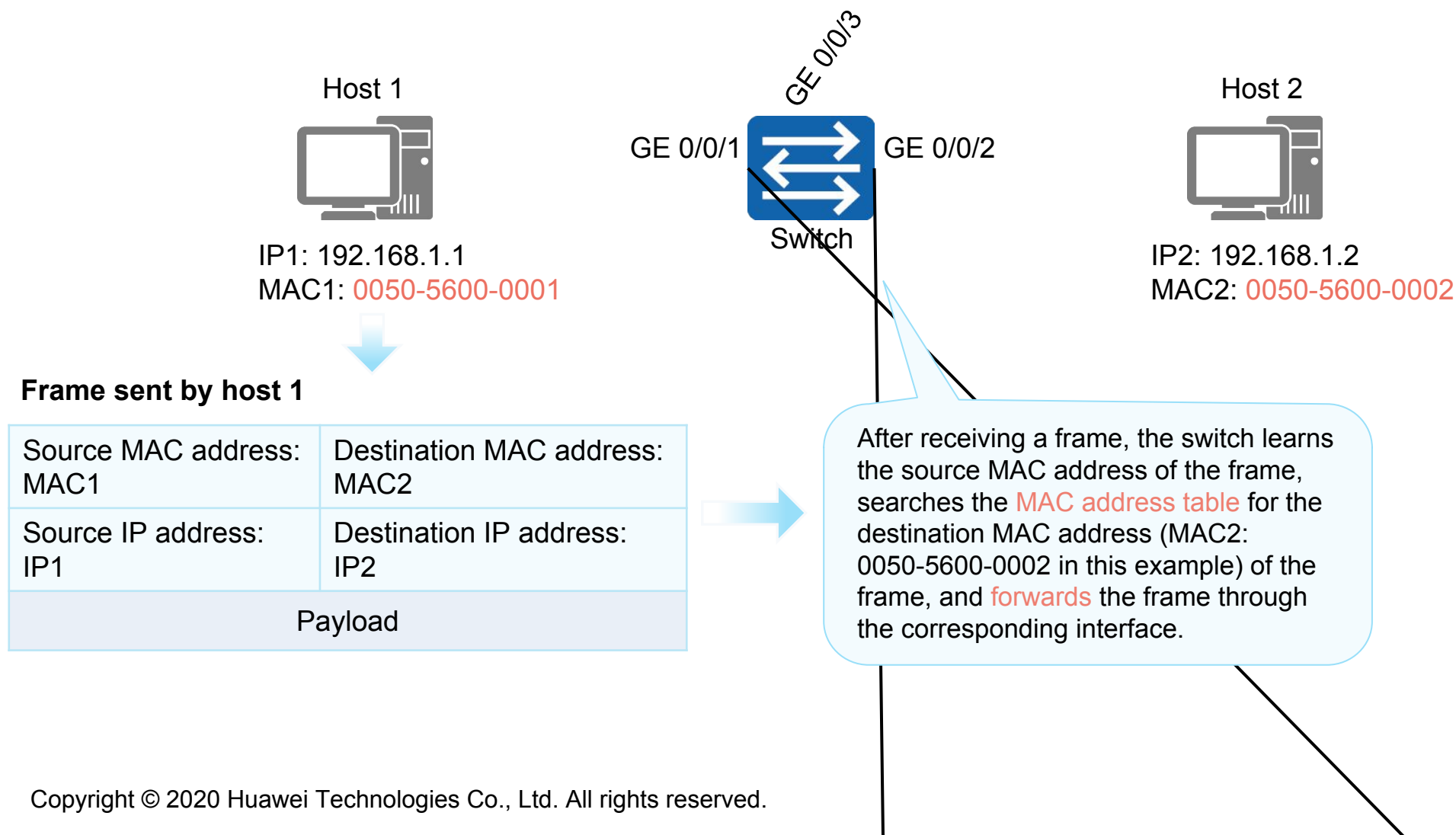
# Layer 2 Ethernet switch

Layer 2 Ethernet switches forward data through Ethernet interfaces and can address and forward data only according to the MAC address in a Layer 2 header (Ethernet frame header).





# Working Principles of Switches

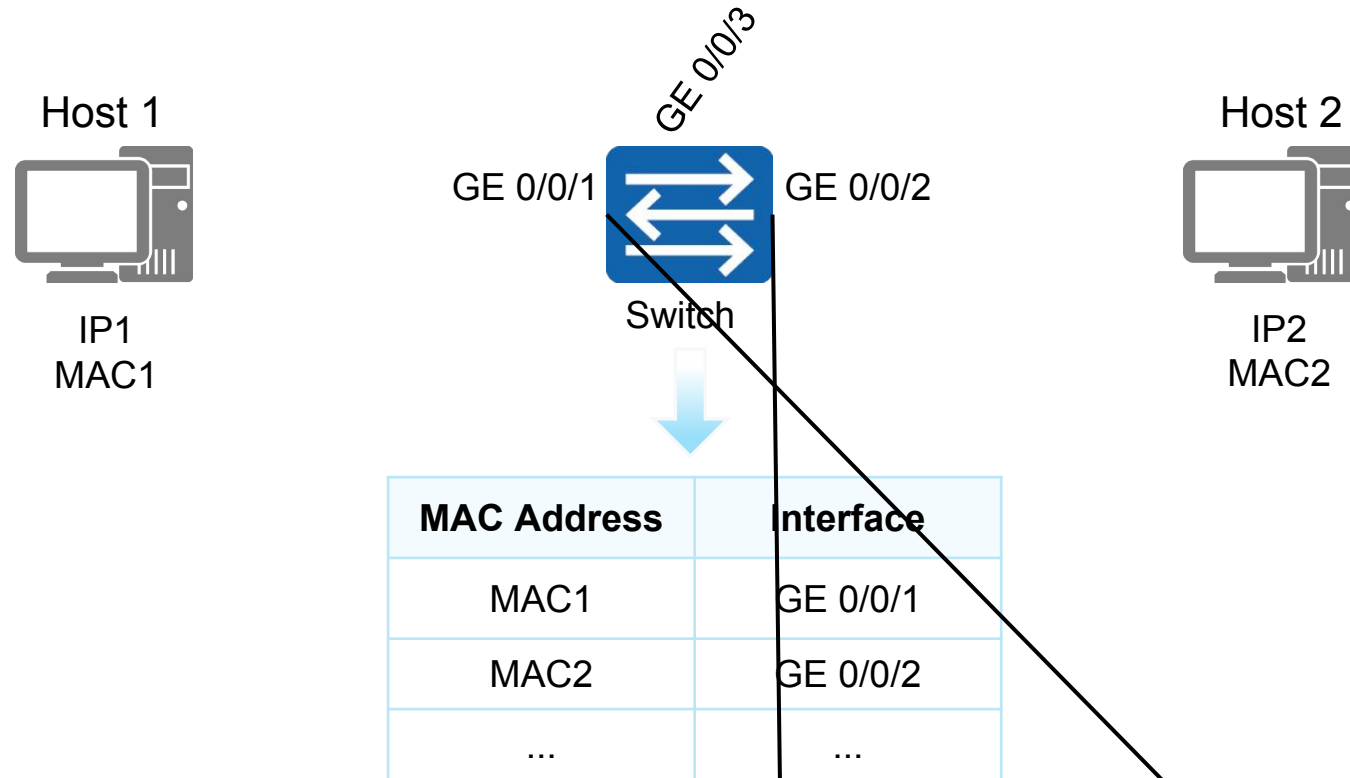






# MAC Address Table

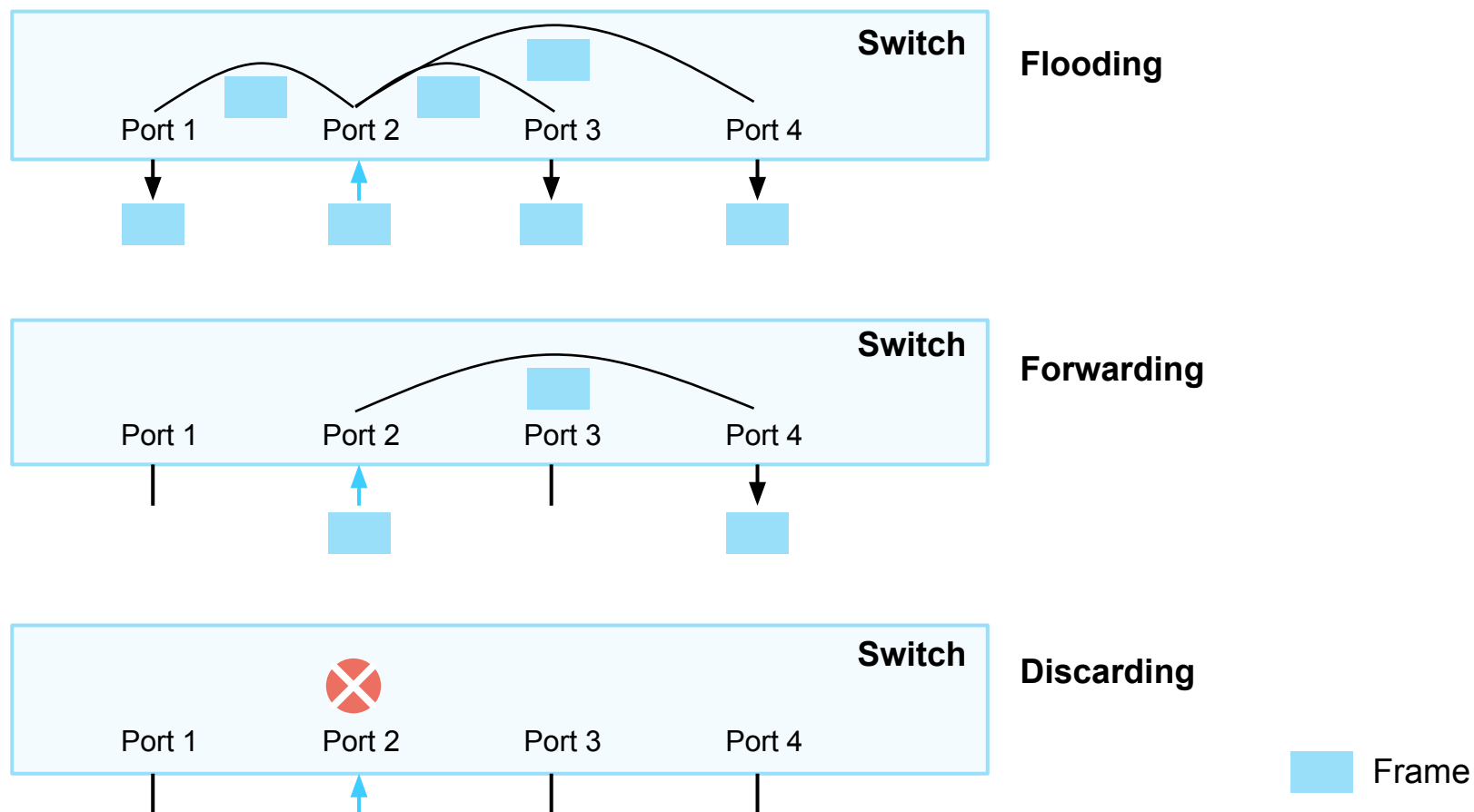
- Each switch has a MAC address table that stores the mapping between MAC addresses and switch interfaces.





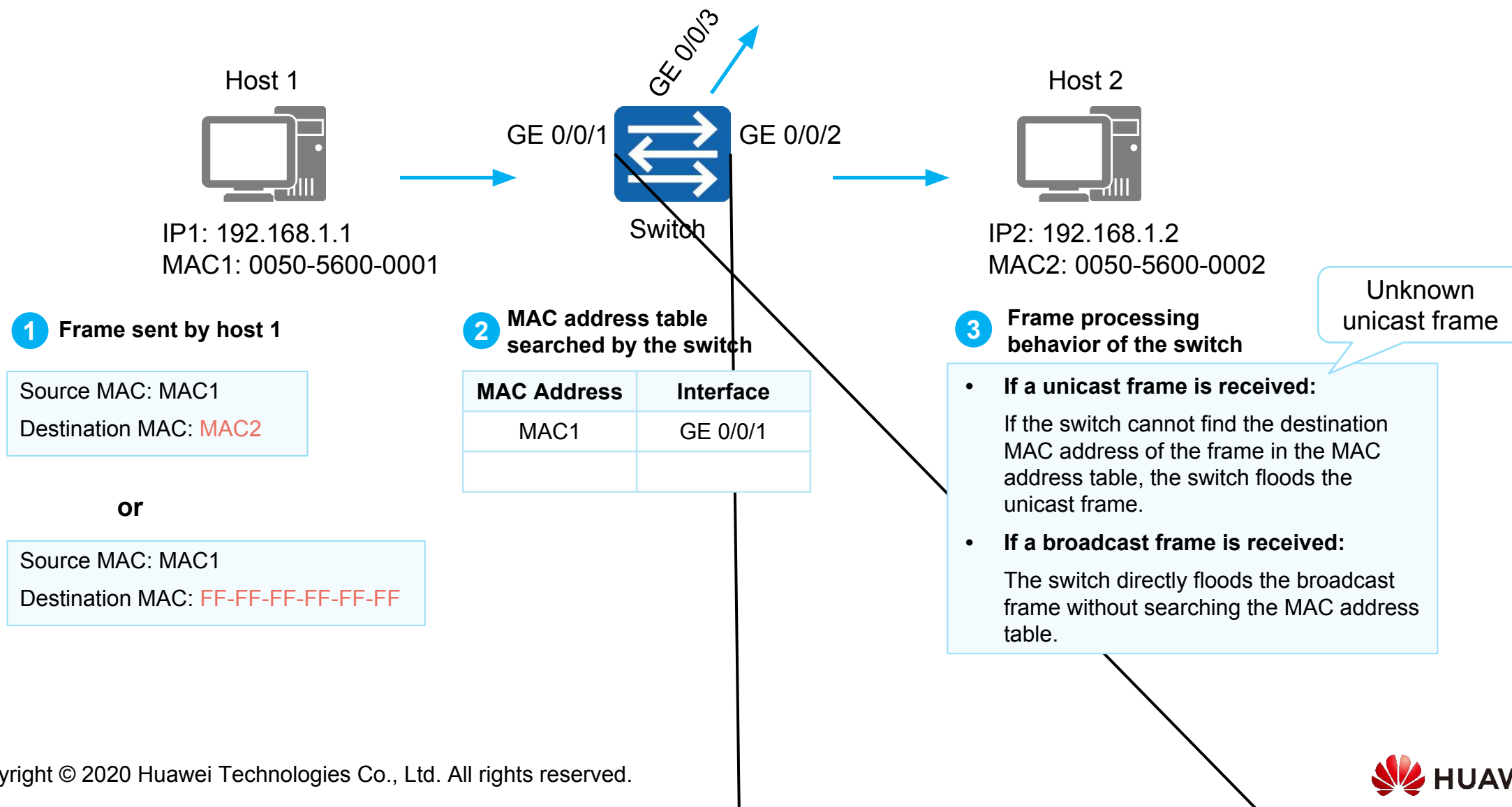
# Three Frame Processing Behaviors of a Switch

- A switch processes the frames entering an interface over a transmission medium in three ways:



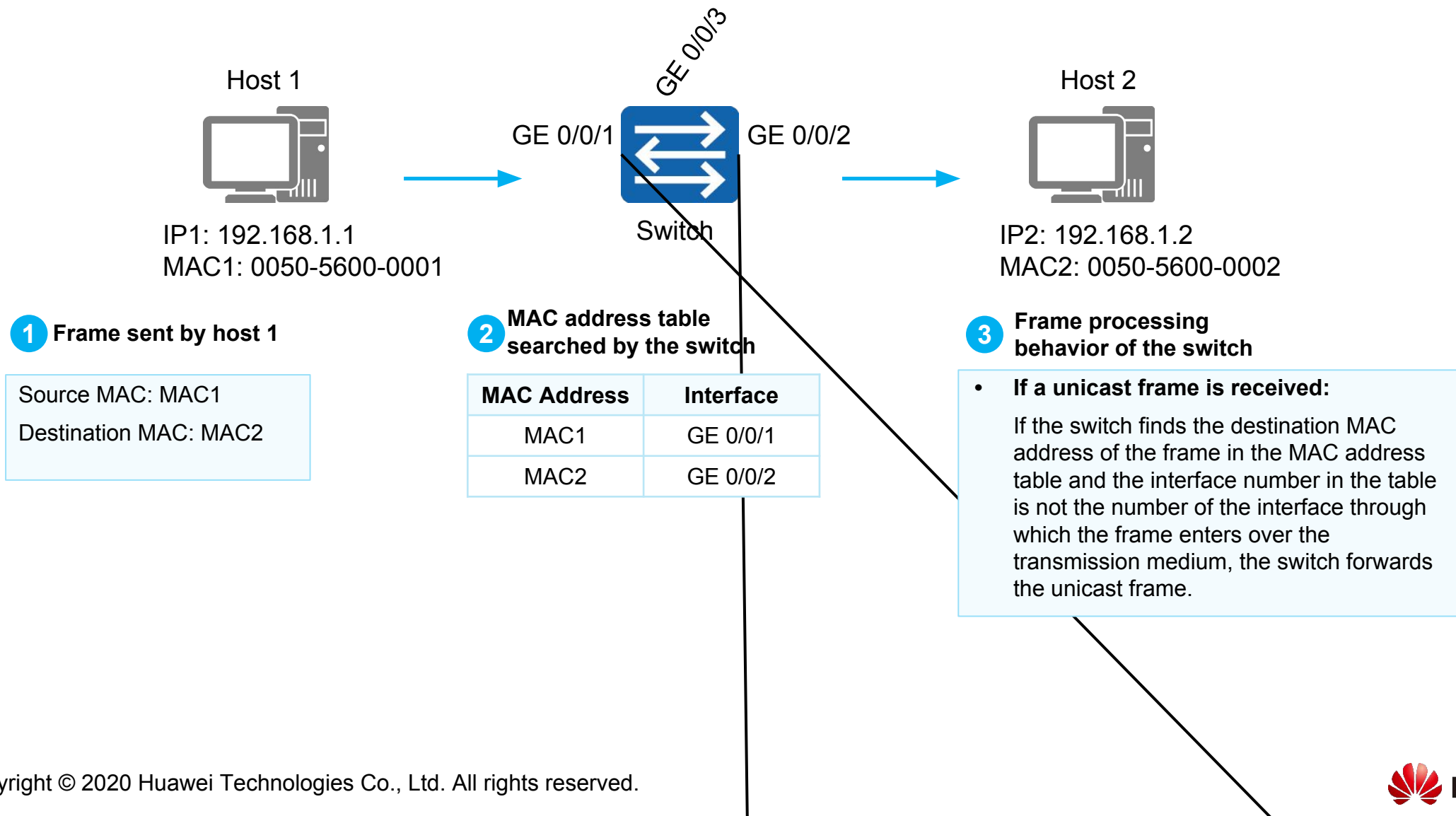


# Flooding



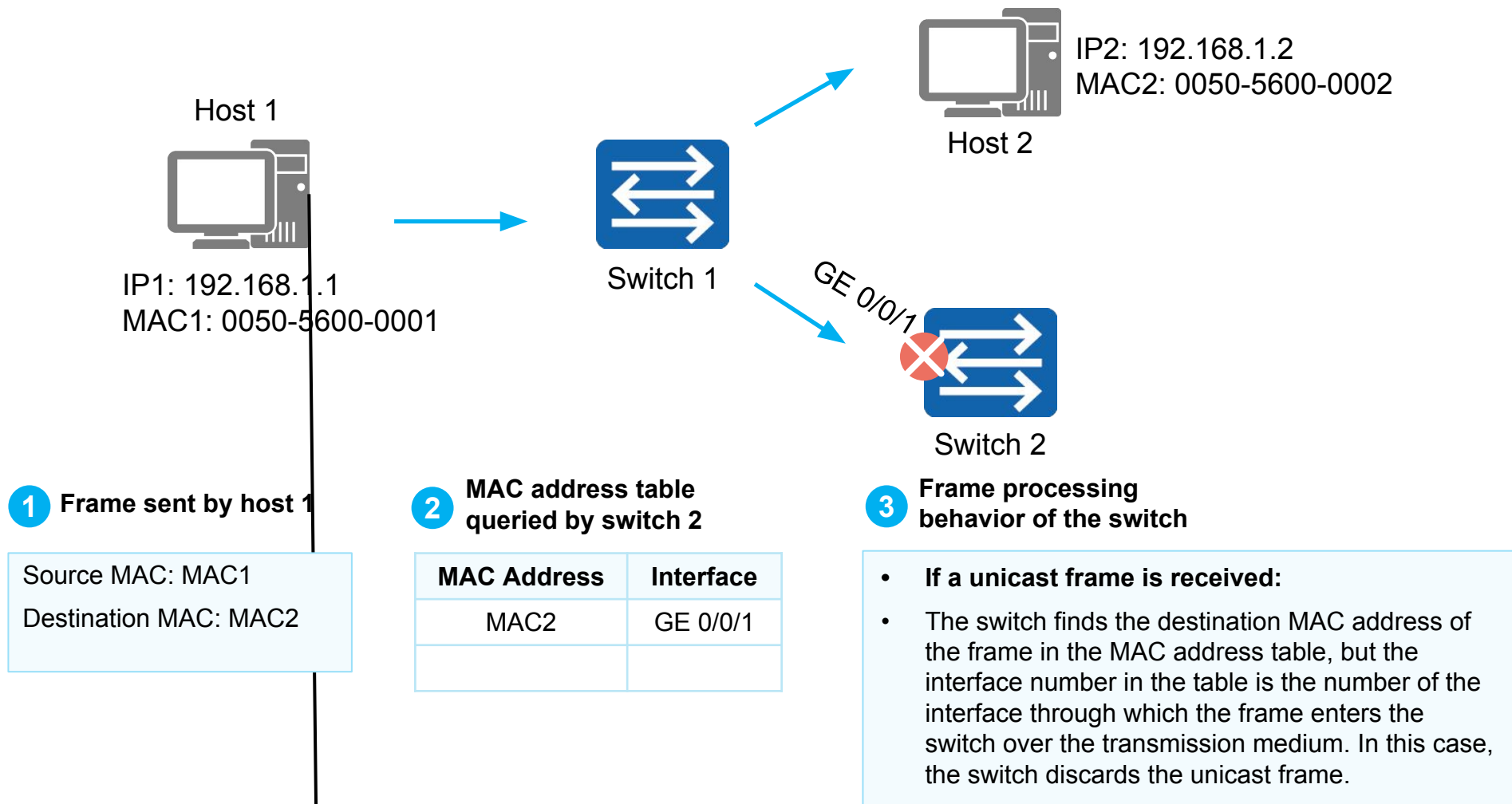


# Forwarding



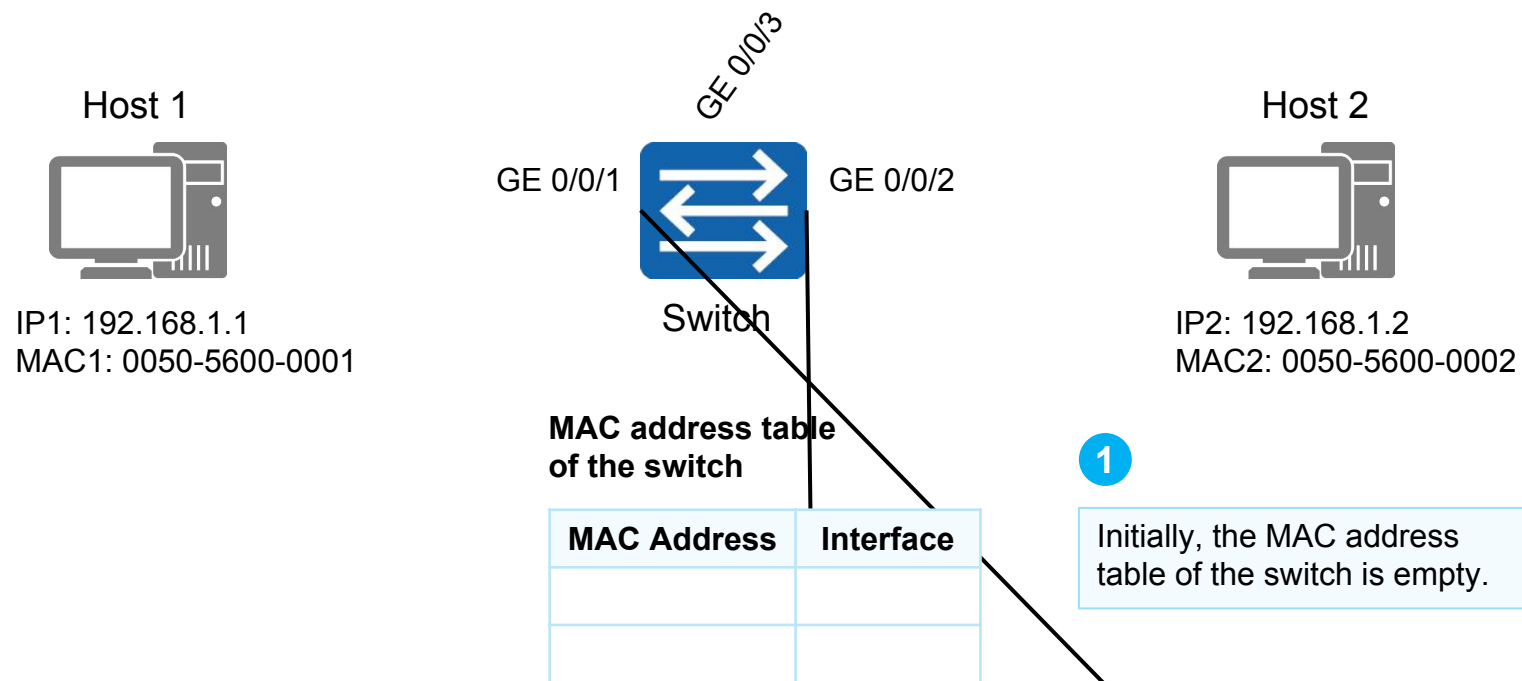


# Discarding



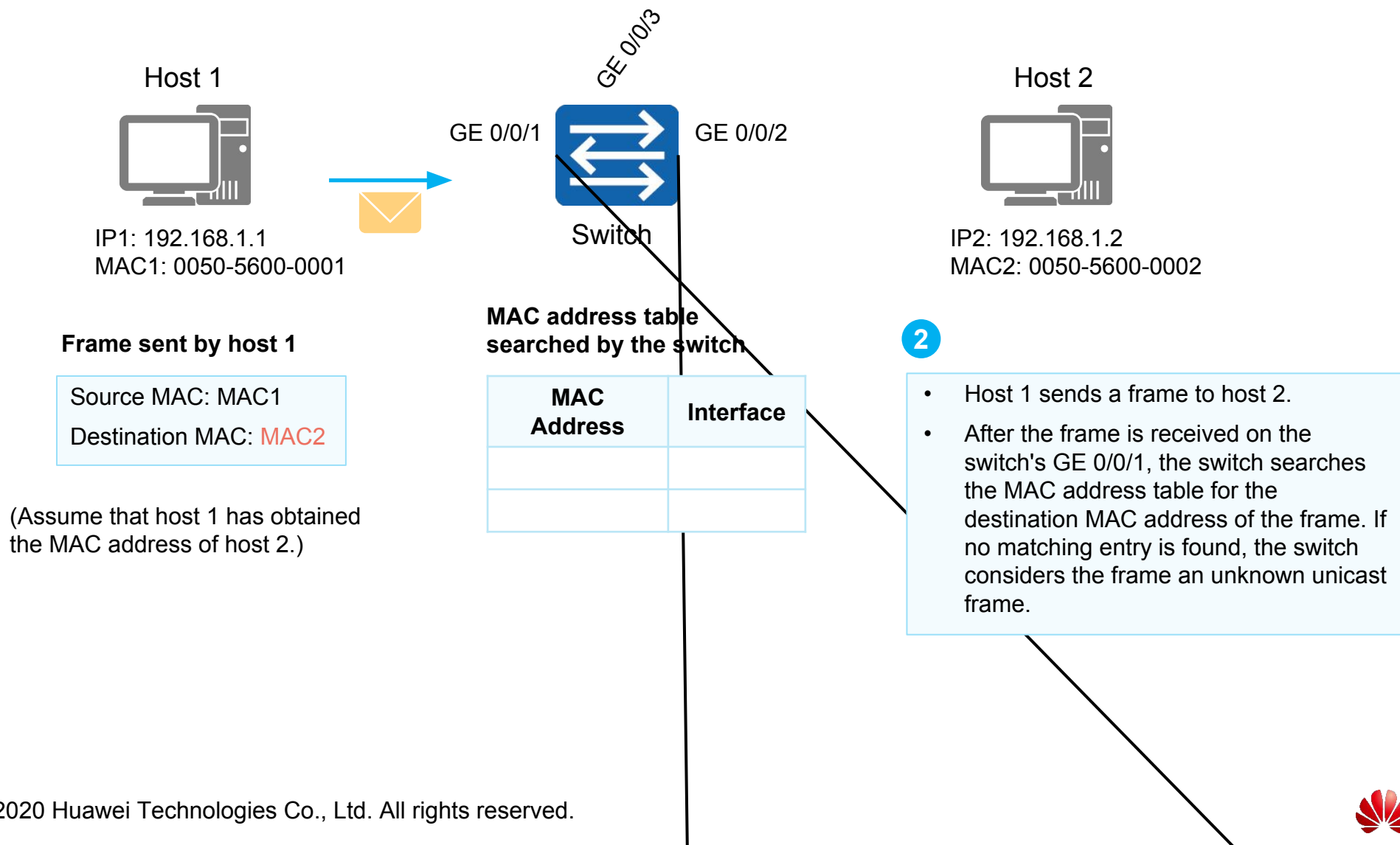


# MAC Address Learning on a Switch (1)



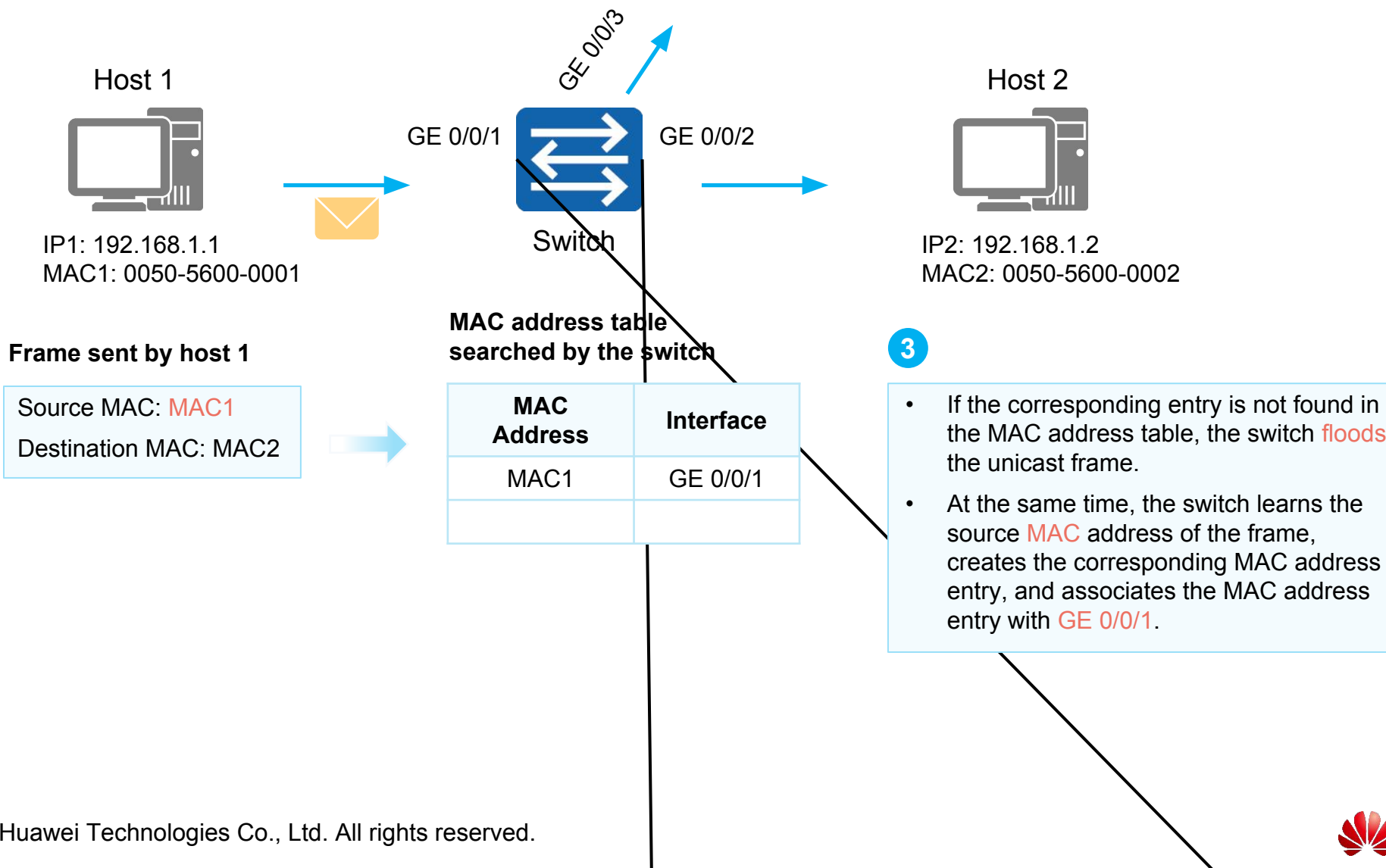


# MAC Address Learning on a Switch (2)





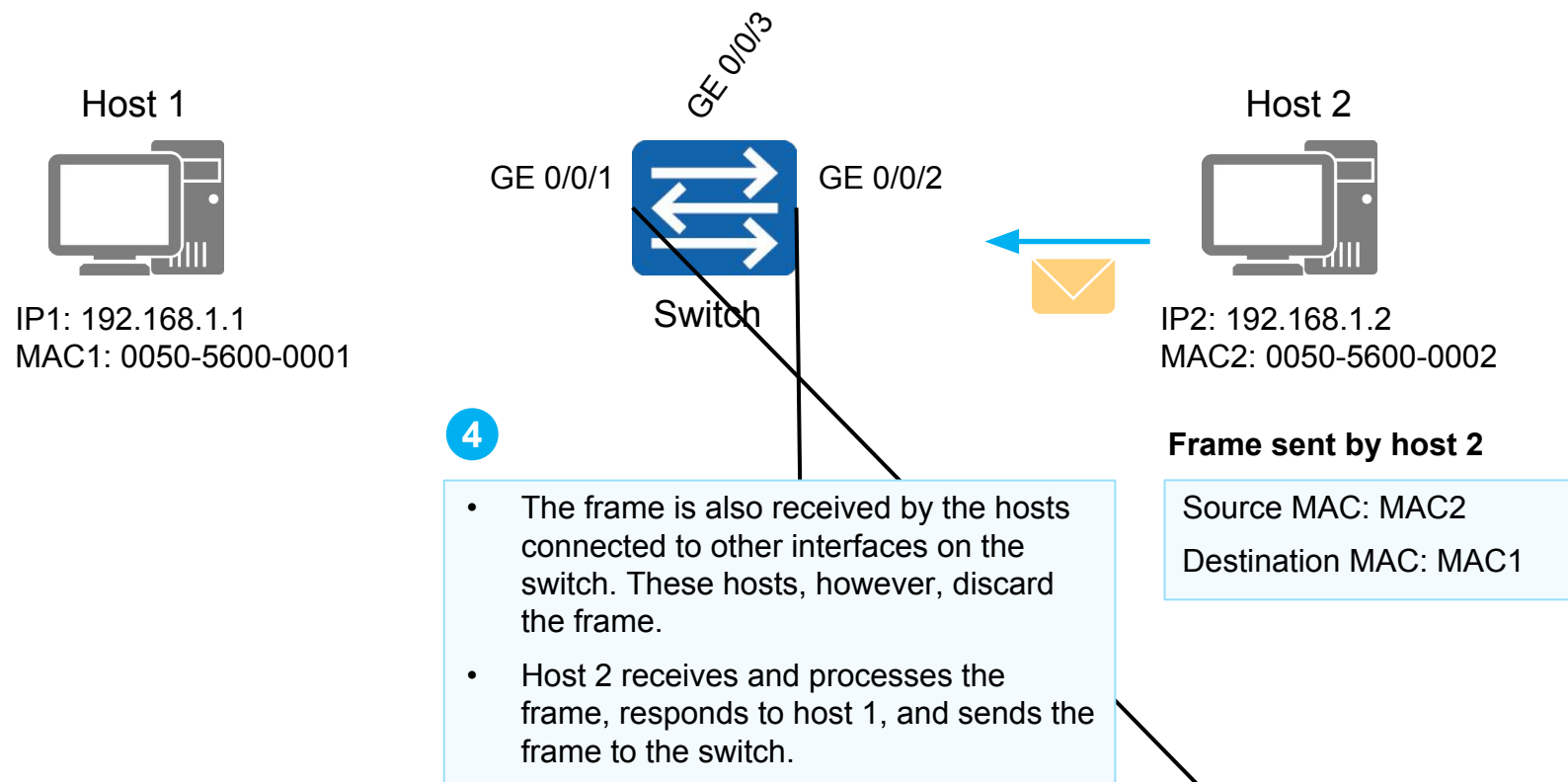
# MAC Address Learning on a Switch (3)





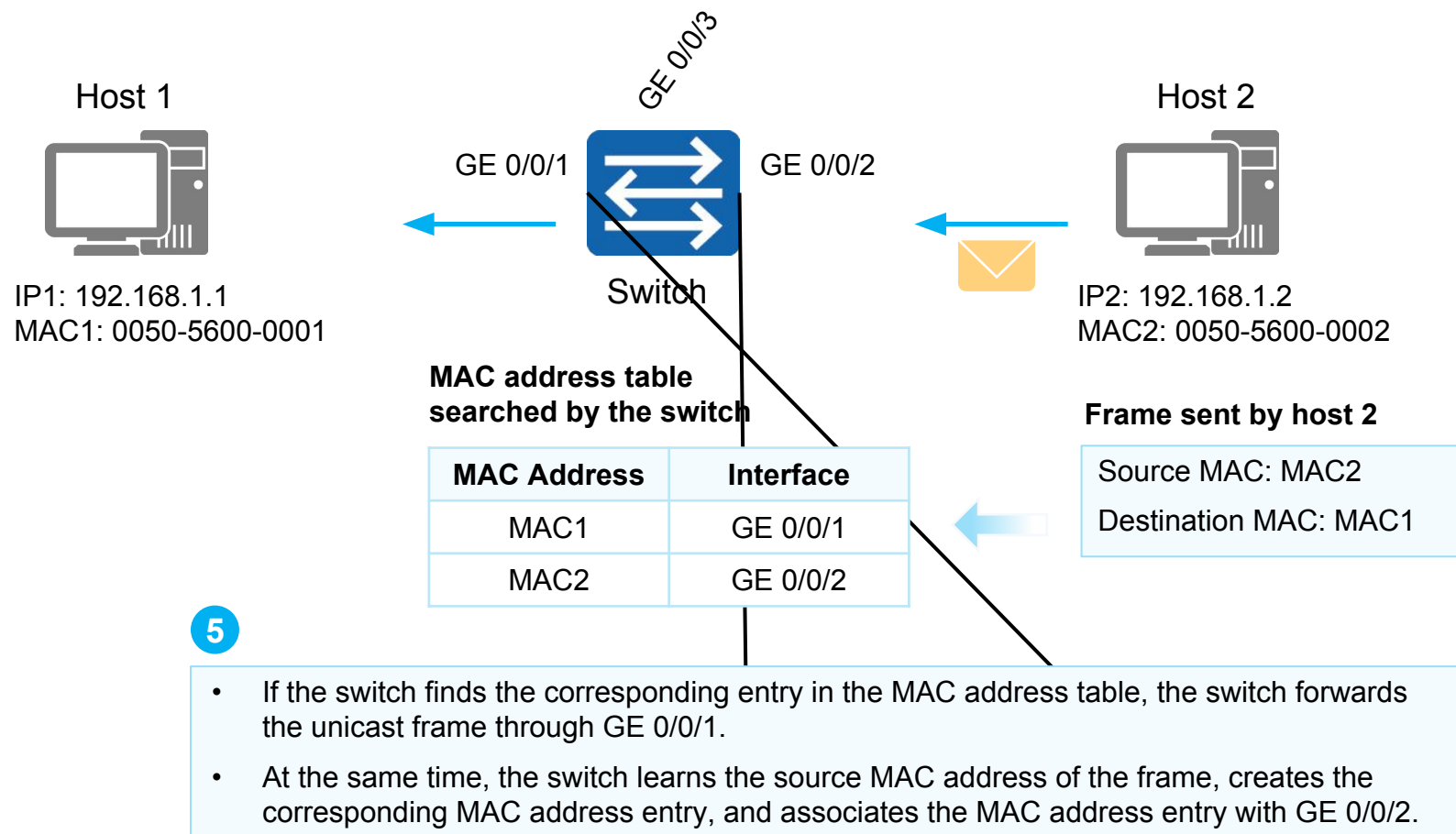


# MAC Address Learning on a Switch (4)





# MAC Address Learning on a Switch (5)





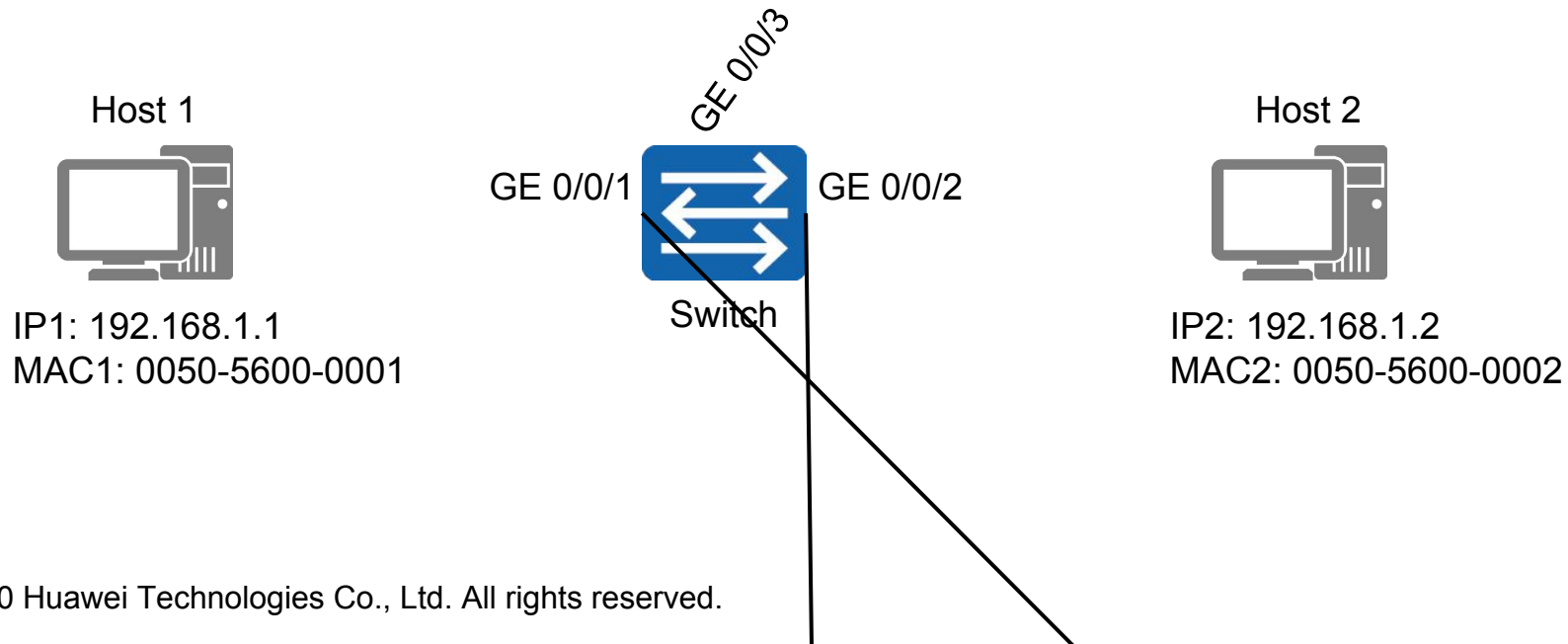
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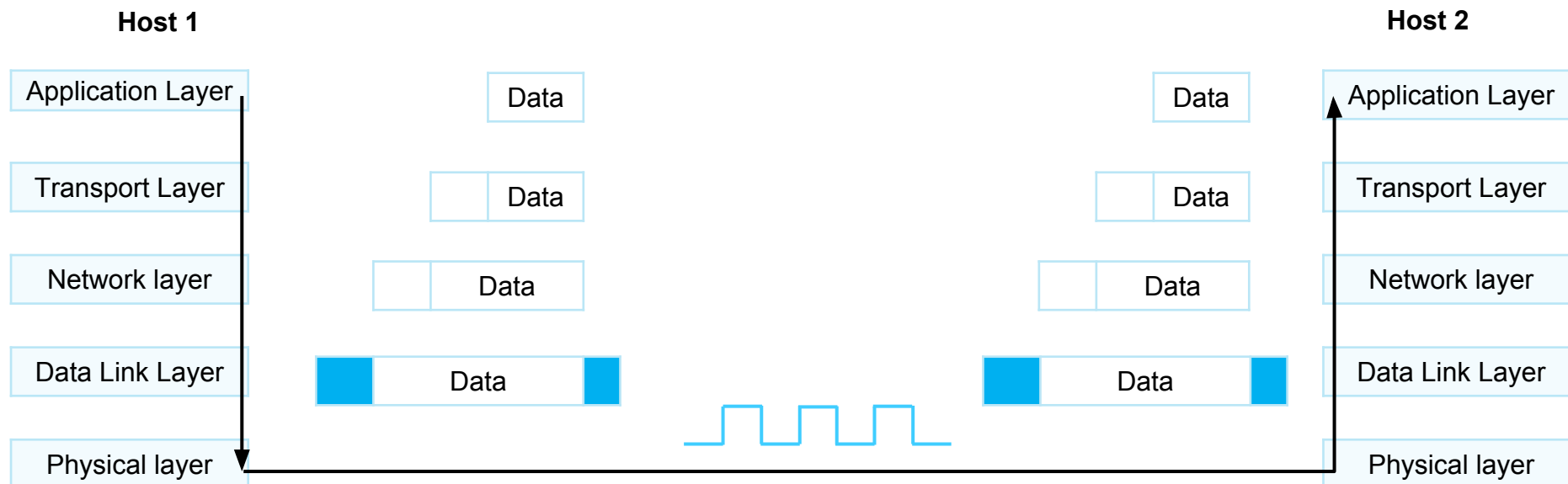
# Process of Data Communication Within a Network Segment

- Scenario description:
  - Task: Host 1 wants to access host 2.
  - Host: The host is in the initialized state and only knows its own IP address and MAC address (assume that the IP address of the peer host has been obtained).
  - Switch: The switch is just powered on and in the initialized state.





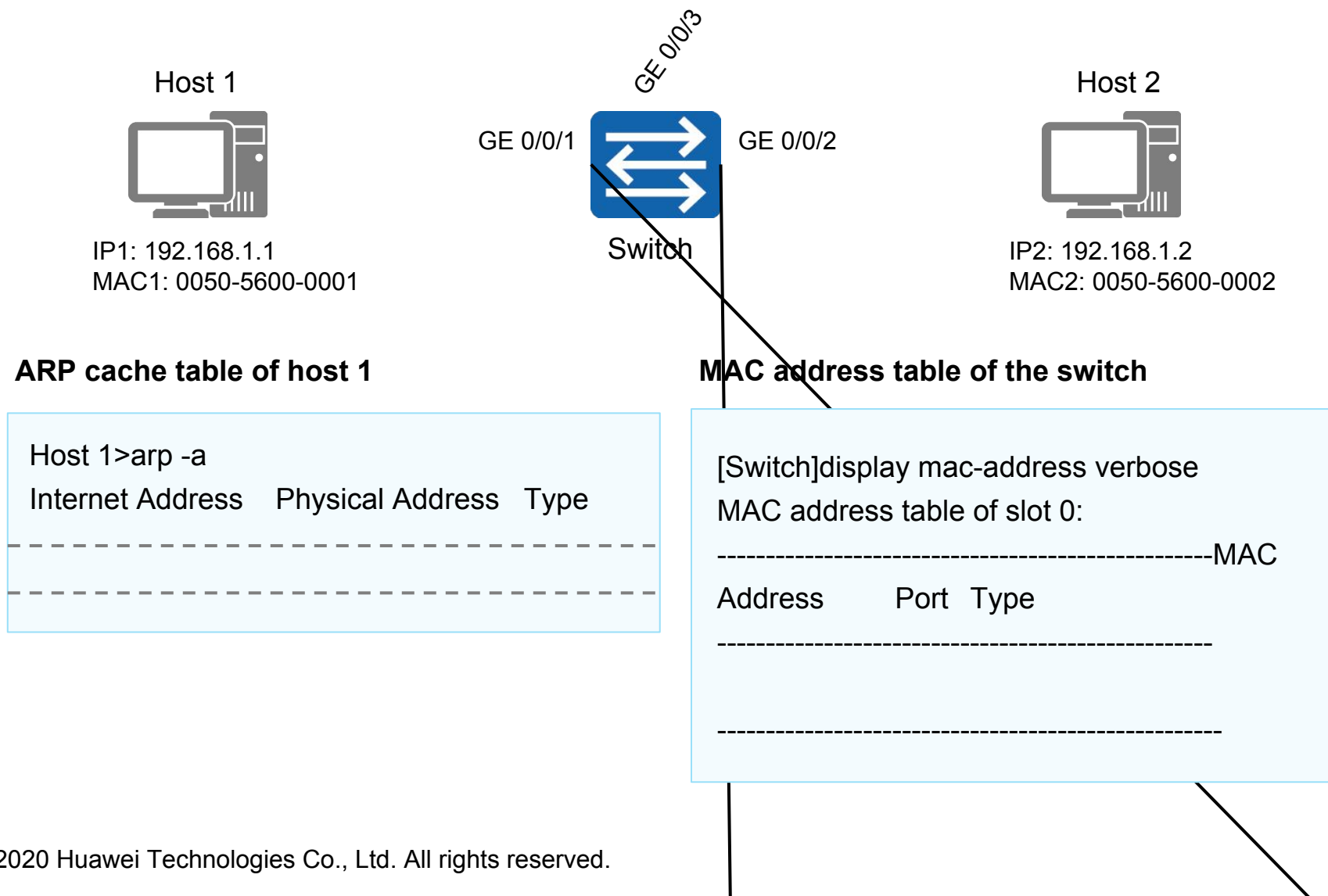
# Data Encapsulation Process



- **Information that needs to be encapsulated:**
- Source MAC address
- Destination MAC address

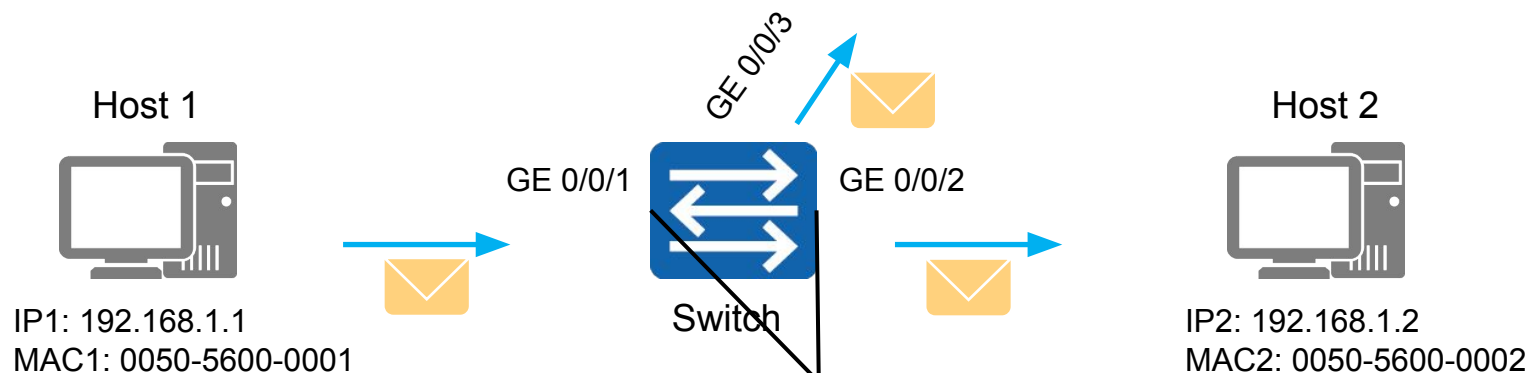


# Initialization





# Flooding Frames



**ARP Request packet sent by host 1**

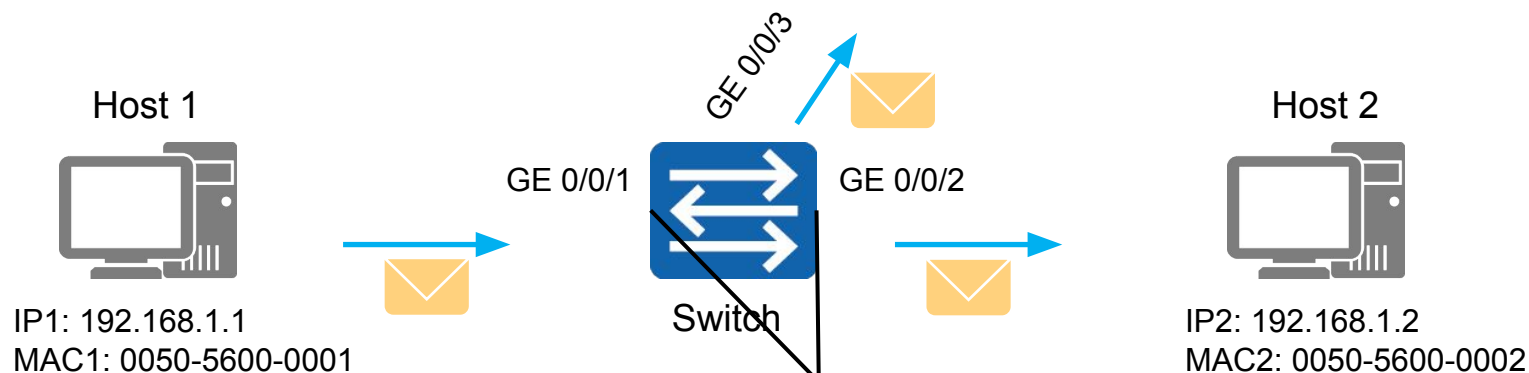
|                                                                                                                                                                    |                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Source MAC address:<br>MAC1                                                                                                                                        | Destination MAC address:<br>FF-FF-FF-FF-FF-FF |
| Source IP address:<br>IP1                                                                                                                                          | Destination IP address: IP2                   |
| Operation type: ARP Request<br>Sender's MAC address: MAC1<br>Sender's IP address: IP1<br>Destination MAC address: 00-00-00-00-00-00<br>Destination IP address: IP2 |                                               |

**MAC address table of the switch**

```
[Switch]display mac-address verbose
MAC address table of slot 0:
-----MAC
Address      Port  Type
-----
-----
```



# MAC Address Learning



## ARP Request packet sent by host 1

|                                                                                                                                                                    |                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Source MAC address:<br>MAC1                                                                                                                                        | Destination MAC address:<br>FF-FF-FF-FF-FF-FF |
| Source IP address:<br>IP1                                                                                                                                          | Destination IP address: IP2                   |
| Operation type: ARP Request<br>Sender's MAC address: MAC1<br>Sender's IP address: IP1<br>Destination MAC address: 00-00-00-00-00-00<br>Destination IP address: IP2 |                                               |

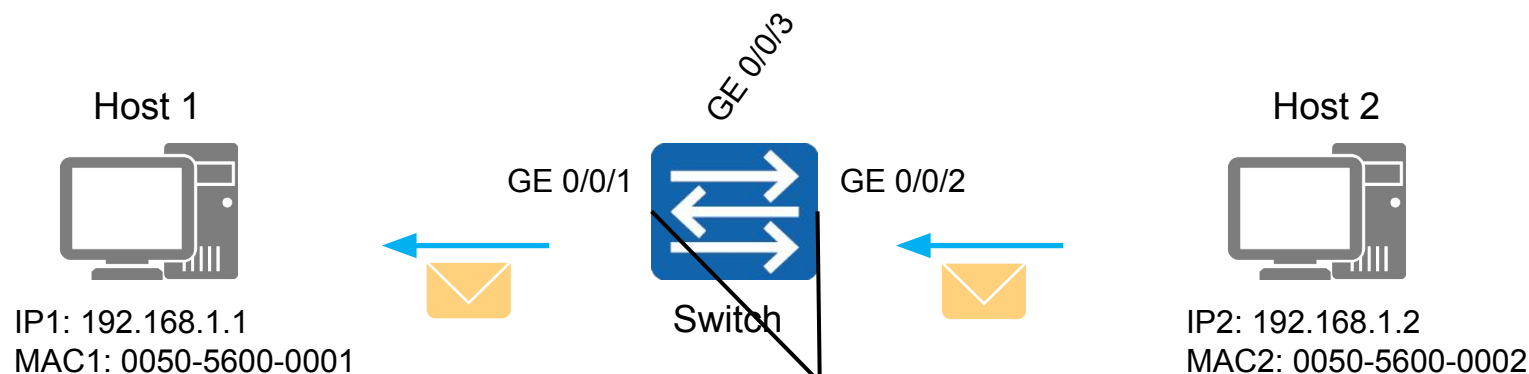
## MAC address table of the switch

```
[Switch]display mac-address verbose
MAC address table of slot 0:
-----MAC
Address      Port  Type
-----0050-5
600-0001    GE0/0/1  dynamic
-----
```





# Reply of the Target Host



MAC address table of the switch

```
[Switch]display mac-address verbose
MAC address table of slot 0:
-----MAC
Address      Port  Type
-----0050-5
600-0001     GE0/0/1  dynamic
0050-5600-0002  GE0/0/2  dynamic
-----
```

ARP Reply packet sent by host 2

|                                                                                                                                                     |                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Source MAC address:<br>MAC2                                                                                                                         | Destination MAC address:<br>MAC1 |
| Source IP address: IP2                                                                                                                              | Destination IP address: IP1      |
| Operation type: ARP Reply<br>Sender's MAC address: MAC2<br>Sender's IP address: IP2<br>Destination MAC address: MAC1<br>Destination IP address: IP1 |                                  |



# Quiz

1. (Single) A Layer 2 Ethernet switch generates a MAC address table entry based on the ( ) of the packet received by an interface.
  - A. Source MAC address
  - B. Destination MAC address
  - C. Source IP address
  - D. Destination IP address.
2. (Single) A switch has eight interfaces. A unicast frame enters the switch through one of the eight interfaces, but the switch cannot find the destination MAC address entry of the frame in the MAC address table. In this case, which of the following operations is performed by the switch? ( )
  - A. Discarding
  - B. Flooding
  - C. Point-to-point forwarding



# Summary

- This section describes the basic information about the Ethernet protocol, Ethernet frame formats, MAC address, and working principles of Layer 2 switches. Specifically, after receiving a frame, a switch learns the source MAC address of the frame and searches the destination MAC address of the frame in the MAC address table. If the destination MAC address exists in the table, the switch forwards the frame through the corresponding interface.
- This course also describes the whole process of data communication within the same network segment based on the working principles of switches.

The background of the image is a blue-tinted photograph of a modern office interior. In the foreground, several groups of business professionals are silhouetted against the bright light coming from large windows. They appear to be in various stages of collaboration, some looking at documents or devices. The floor is highly reflective, mirroring the figures and the light. In the background, through the glass walls, a dense urban skyline with various skyscrapers is visible under a clear sky. The overall color palette is dominated by shades of blue, creating a professional and high-tech atmosphere.

Thank You  
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